

M.S. THESIS DEFENSE

Toward Passwordless Document Encryption

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Overview

- | | | | |
|---|-------------------|--|----------------|
| 1 | Problem | Sensitive documents, MFT breaches at scale, and why current protection fails. | <i>Ch. 1-2</i> |
| 2 | Background | Encryption & passwordless authentication building blocks.
Framework and evaluation showing no current method provides all benefits. | <i>Ch. 2-4</i> |
| 3 | Design | Novel system: age + WebAuthn PRF together in a working prototype | <i>Ch. 5</i> |
| 4 | Study | Counterbalanced within-subjects evaluation of the new system
against Adobe Acrobat (N = 21). | <i>Ch. 6</i> |
| 5 | Results | Usability, acceptance, and adoption findings. Takeaways and future research. | <i>Ch. 7-8</i> |

Research questions

R1

Current Limitations

What are the security, usability, and deployability limitations of current document encryption methods?

R2

Novel Design

How can open-source tools be used to design a passwordless, non-proprietary, per-file document protection system?

R3

Usability + User Acceptance

What is the usability and user acceptance of a passwordless document encryption system?

SECTION 1 / 5

Problem

Sensitive documents, MFT breaches at scale, and why current protection fails.



1 Problem

2 Background

3 Design

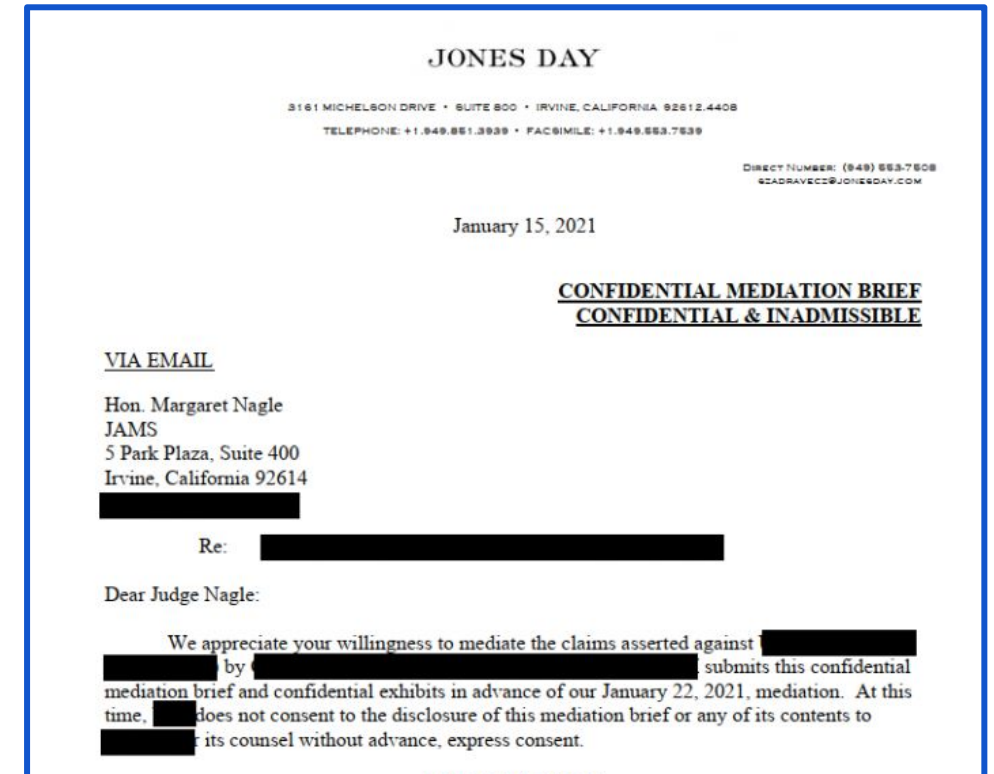
4 Study

5 Results

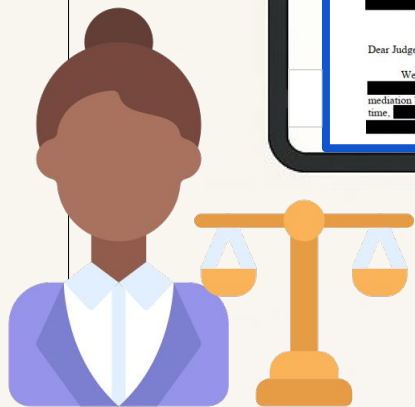
Sensitive documents are everywhere

Finance, legal, healthcare,
government.

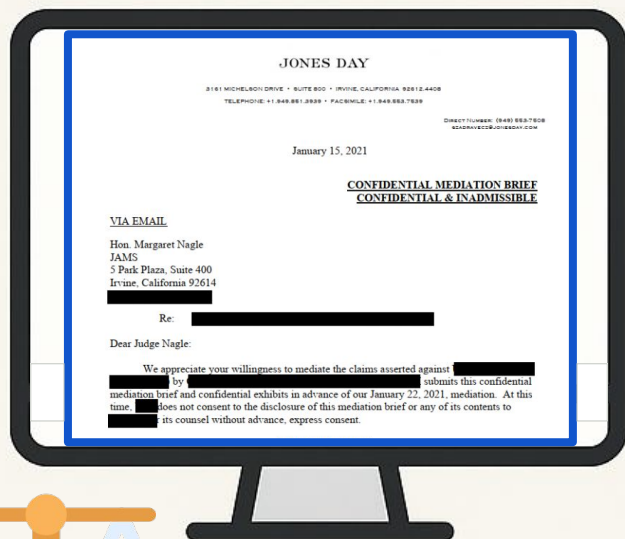
Confidential files are created everyday and must be shared with external users outside any single organization's control.



Example: confidential legal brief



Lawyer



Judge

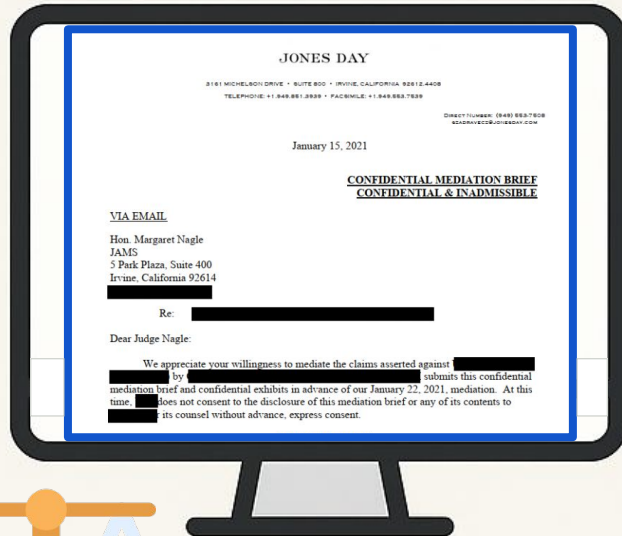
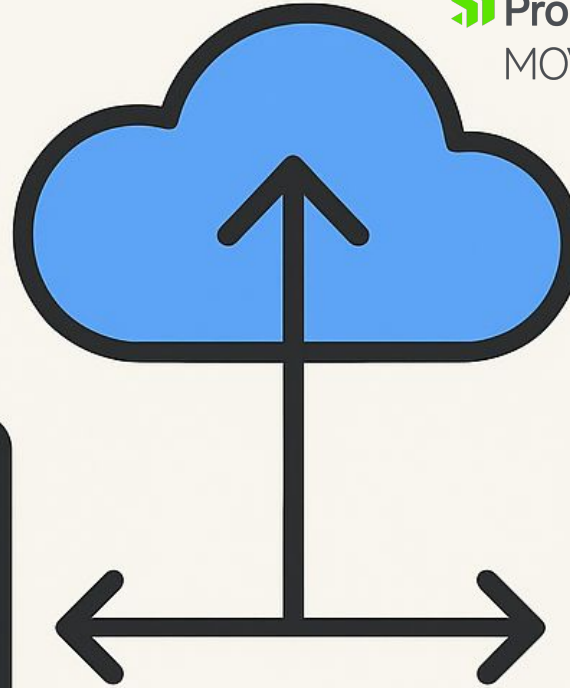


MFT

sharetru

FORTRA[™] GoAnywhere **Cleo**

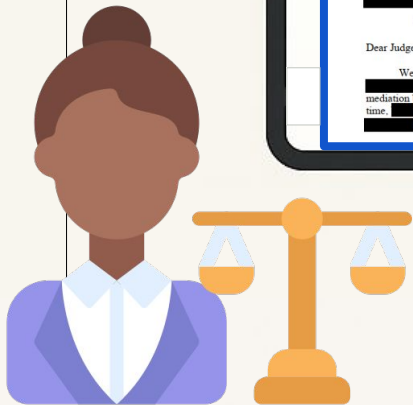
Progress[®]
MOVEit[®]



Lawyer



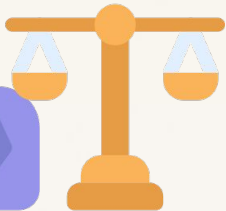
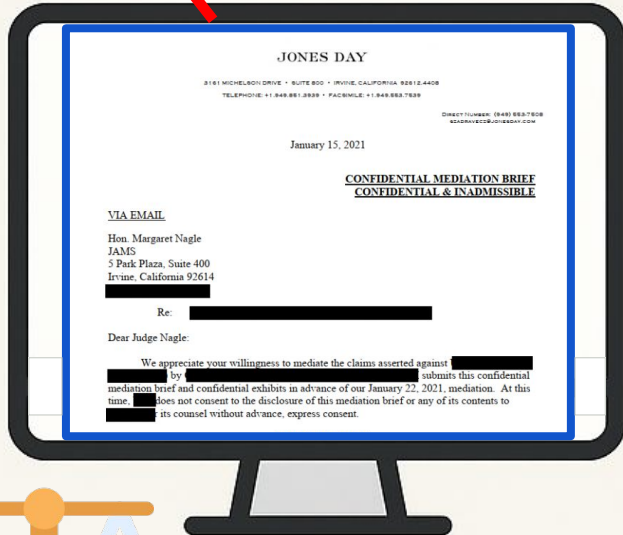
Judge



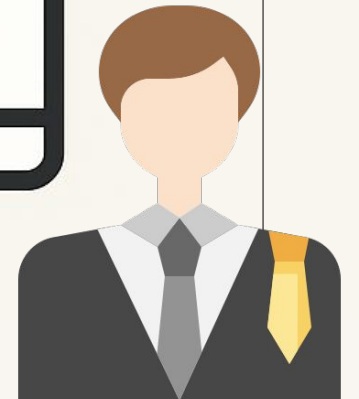


MFT

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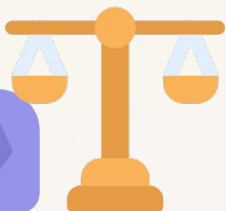
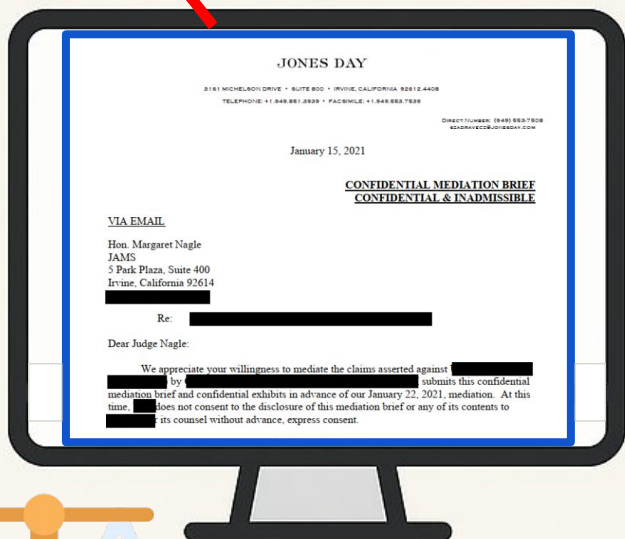
Lawyer



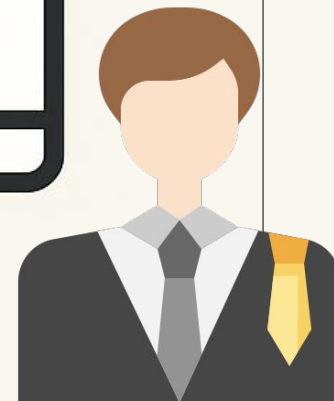
Judge



MFT



Lawyer



Judge



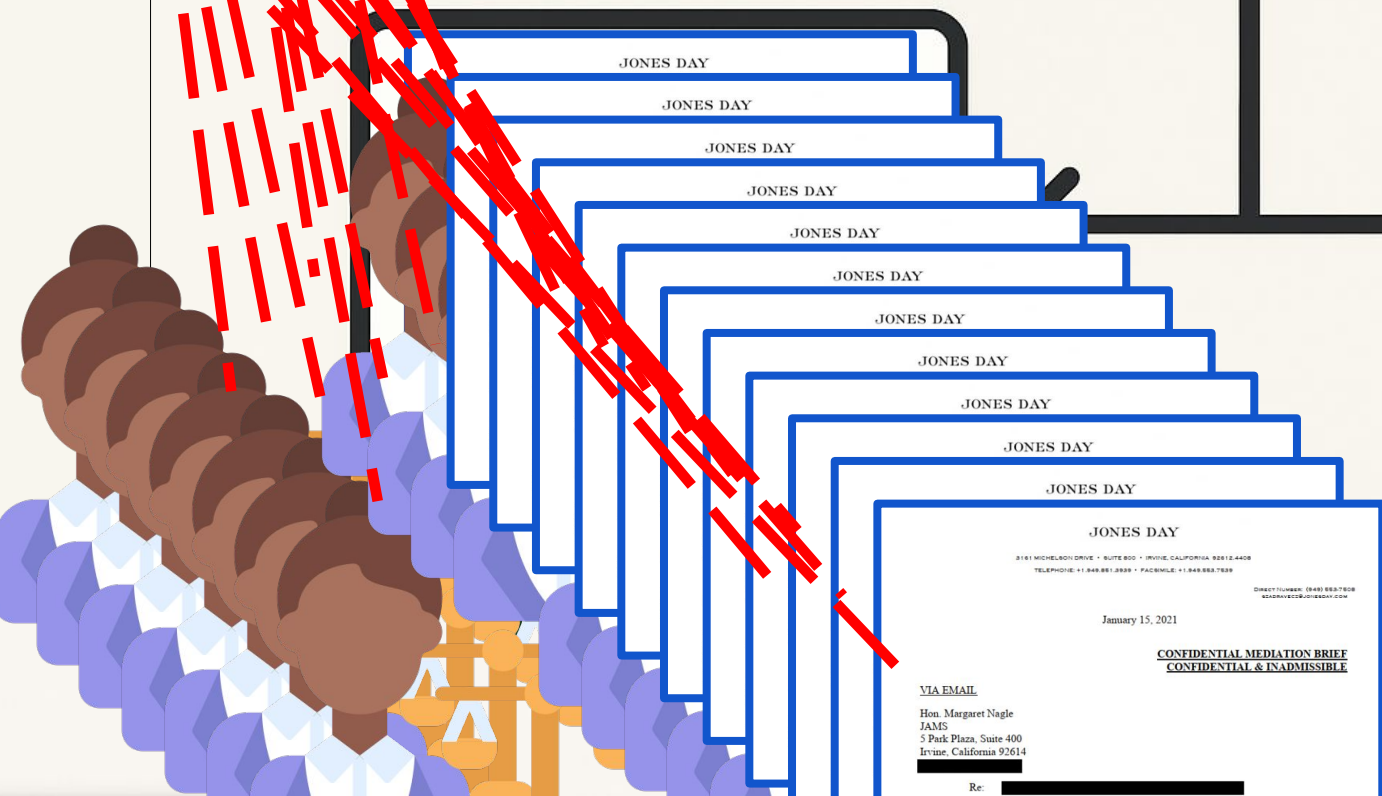


MFT

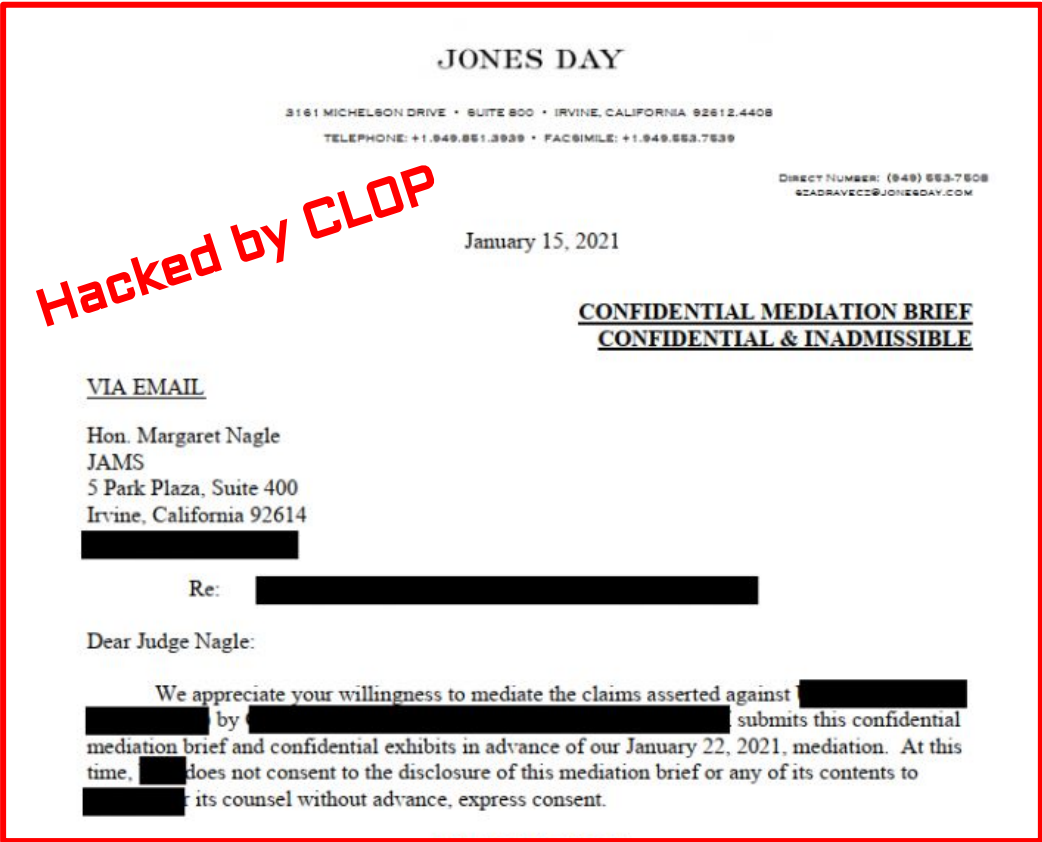
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Judge



This is not hypothetical.



May 2023
MOVEit breach

x 95,788,491
people

28% of 

Organization	Impacted Individuals	Industry
Welltok	14.76 million	Health care
Maximus	11.3 million	Government IT Contractor
Delta Dental of California and affiliates	6.9 million	Dental Insurance
Louisiana Office of Motor Vehicles	6 million	Government
Alogent	4.5 million	Banking

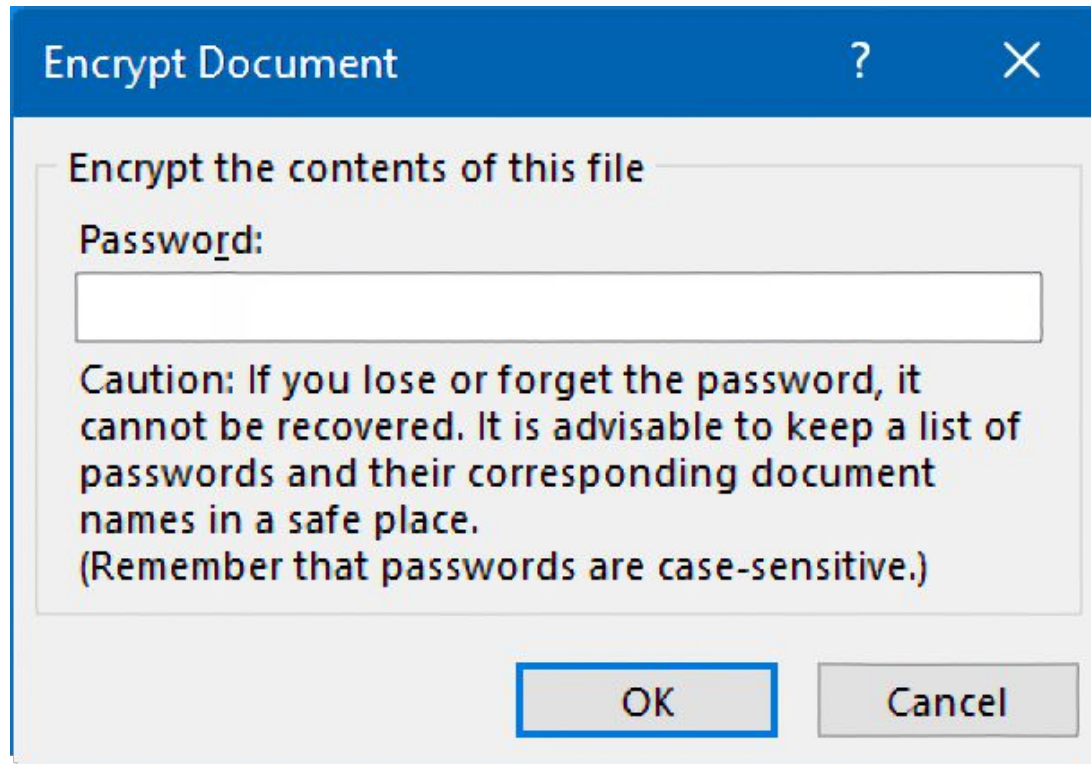
A systematic pattern of Managed File Transfer (MFT) breaches:

● Accellion	Dec 2020
● Serv-U	Nov 2021
● GoAnywhere	Jan 2023
● MOVEit	May 2023
● Cleo	Dec 2024
● Wing FTP	Jul 2025

Adversary Threat Model

	Threat	Design Requirement	Benefit(s)
I	Bulk server-side file access	Granular per-document encryption, no shared encryption key	S4 Granular-Protection
II	Brute-force password cracking post-exfiltration	Passwordless key derivation	S2 Resilient-to-Guessing U1 Memorywise-Effortless
III	Unauthorized redistribution of exfiltrated files	Recipient-bound encryption	S3 Resilient-to-Third-Party-Leak
IV	Server compromise exposing encryption keys	Non-custodial, client-side key derivation, zero-knowledge server	S3 Resilient-to-Third-Party-Leak

Usability...



Two things are broken



Perimeter & at-rest only

MFT and cloud storage encrypt files only “at rest.” Once an attacker is inside the system, every file can be read.

| One breach → all documents exposed



Password-based encryption

Word / Acrobat passwords are vulnerable to brute force, can't be recovered if lost, and the password must somehow be shared securely.

| Stuck on 1990s technology

SECTION 2 / 5

Background

Encryption and passwordless authentication building blocks and a framework and evaluation showing no current method provides all benefits.



1 Problem

2 Background

3 Design

4 Study

5 Results

A framework of 15 benefits

Usability (5)

- Memorywise-effortless
- Scalable for users
- Easy to learn
- Easy recovery from loss
- Seamless sharing

Deployability (6)

- Negligible cost per user
- Negligible IT support cost
- Compatible
- Mature
- Non-proprietary
- Guest-friendly

Security (4)

- Resilient to observation
- Resilient to guessing
- Resilient to third-party leak
- Recipient-bound

No existing method does it all

Auth. Type	Scope	License	Examples	Usability					Deployability						Security			
				U1	U2	U3	U4	U5	D1	D2	D3	D4	D5	D6	S1	S2	S3	S4
Password-based	File level	Incumbent	PDF, Microsoft Word															
		Proprietary	WinRAR, AxCrypt															
		Open Source	7-zip, Picocrypt															
Password-based	Full disk	Proprietary	BitLocker, FileVault															
		Open Source	VeraCrypt, LUKS															
Cloud-based	File share	Proprietary	MFT, SharePoint, Box															
		Open Source	CryptPad, Nextcloud, SFTP															
Passwordless	File level	Proprietary	PreVeil, Virtru															
		Open Source	age, GPG															

Satisfies Partially Satisfies Does Not Satisfy

Sept 2025, Doc Eng Nottingham, UK





Document Encryption in Practice: A Comparative Framework and Evaluation

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ABSTRACT
Document files with sensitive information are used across nearly every industry. In recent years, cyberattacks have resulted in millions of sensitive documents being exposed. Although document encryption methods exist, they are often flawed in terms of usability, security, or deployability. We present a structured framework for evaluating document encryption methods, adapting the usability-deployability-security (UDS) model to the document encryption context. We apply this framework to compare current methods, performing a comprehensive evaluation of nine document protection methods, including password-based, passwordless, and cloud-based approaches. Our analysis across 12 design properties highlights the benefits and limitations of current methods. We propose strategies and design recommendations to address key limitations such as memory-wise effort, granular protection, and shareability.

CCS CONCEPTS
• Security and privacy → Software and application security.

KEYWORDS
Document encryption, Secure document engineering, Security and usability

ACM Reference Format:
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1 INTRODUCTION
Electronic documents containing highly sensitive data are nearly ubiquitous, used across a wide range of industries. Examples include health records, legal counsel, financial data, trade secrets, and employment contracts. These sensitive documents are often stored and shared using cloud services like managed file transfer (MFT) applications. Users trust these platforms to keep their data secure. However, in recent years, several large MFT applications experienced mass exploit events that compromised millions of confidential documents, including personal health data, government records, and legal files [6]. The largest incident resulted in the exposure of 95,788,491 individuals' personal information [12]. Although MFT applications encrypt files 'at rest', all documents are unencrypted and readable by a privileged user session. From 2020 to 2023, five vulnerabilities allowing remote privileged access to MFTs were discovered and abused, resulting in major data breaches¹ [4]. These incidents underscore ongoing challenges in the state of document encryption. While several approaches exist—such as password-based, proprietary, and storage provider encryption—each comes with major limitations. Issues of security, usability, and deployability hinder widespread adoption and reduce effectiveness. Improved methods are needed to address these challenges.

In this paper, an analytical framework for measuring document protection methods is presented consistent with approaches taken to evaluate authentication schemes in related domains [1–3, 5]. This framework provides a semi-structured evaluation methodology and benchmark for future document protection methods [3]. We aim to answer the following question: *What usability, deployability, and security limitations exist in current document encryption methods?*

2 RELATED WORK
Several research efforts have demonstrated weaknesses with current file security measures, including digital signatures [10], XML encryption, and PDF encryption [9]. Work has been done on document integrity and non-repudiation both to improve the integrity of documents [11] and to uncover vulnerabilities in integrity features such as PDF certification signatures [10]. This paper focuses on the challenge of maintaining confidentiality instead of integrity or non-repudiation. Little research has been done to directly assess the usability of document encryption methods. However, extensive usability research has been performed on passwords [1, 13], passwordless technology [1], and encrypted email [5] which each present important connections to the usability challenges of document encryption.

3 FRAMEWORK
The usability-deployability-security (UDS) framework was first presented by Beaune et al. [5] to evaluate password replacement schemes and has subsequently been reused [1, 2], expanded [13], and adapted [5]. The methodology draws on usability evaluation methods such as feature inspection and Nielsen's heuristic analysis to identify common design properties and develop a taxonomy comparing the benefits offered by each method. The focus is on

¹Academy Dec 2020, ServU Nov 2021, Cyberwatch Jan 2023, MOVEit May 2023, and Citrix Dec 2024.

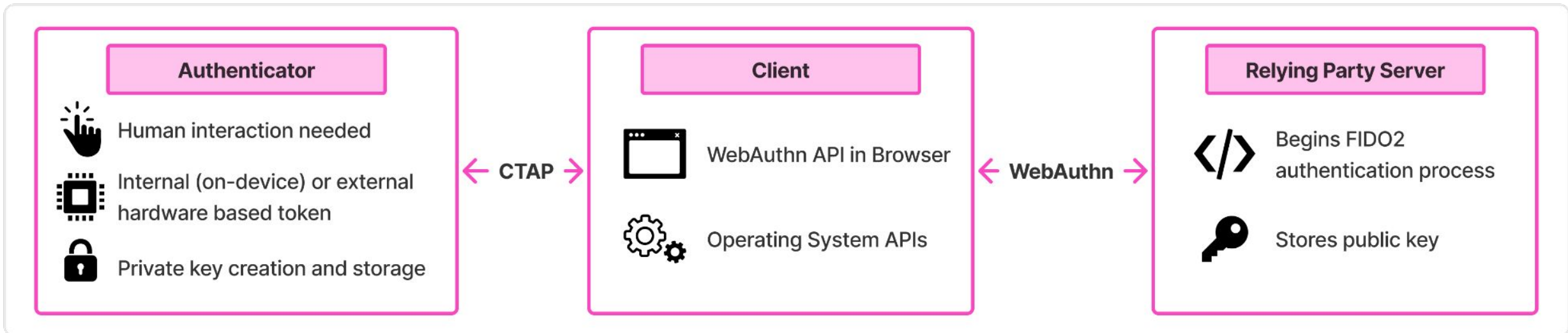
Building block 1: age encryption



age: open-source file encryption

- ✓ **Per-file keys** — A unique 128-bit key wraps each individual file, not one shared secret.
- ✓ **Multi-recipient** — A file can be encrypted to several recipients, each decrypting independently.
- ✓ **Post-quantum ready** — Native hybrid ML-KEM-768 recipients (age 1.3.0) for cryptographic agility.
- ✓ **Open-source** — Simple format, no licensing fees. Satisfies the non-proprietary requirement.

Building block 2: FIDO2 (passkeys) + WebAuthn



FIDO2 (passkeys) have reached widespread adoption

Backed by Google, Microsoft, and Apple. Now supported across browsers, operating systems, and major websites.

5 billion

active passkeys in use globally

90%

of consumers are familiar with
passkeys

75%

have enabled passkeys on
some accounts

68%

of organizations deploying or
piloting passkeys

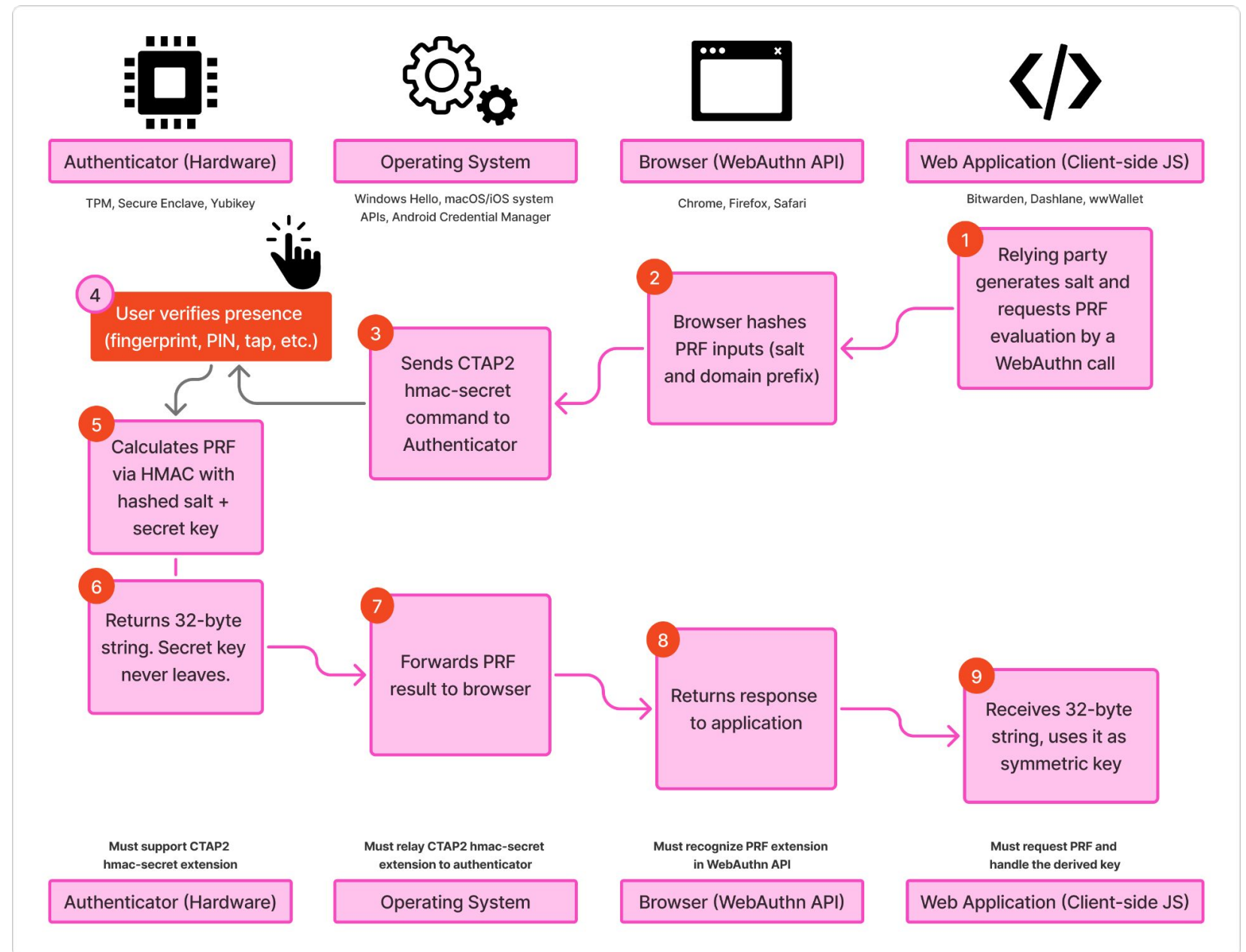
Source: FIDO Alliance, *State of Passkeys 2026* — survey of 11,000 consumers across ten countries.

Building block 3: WebAuthn PRF

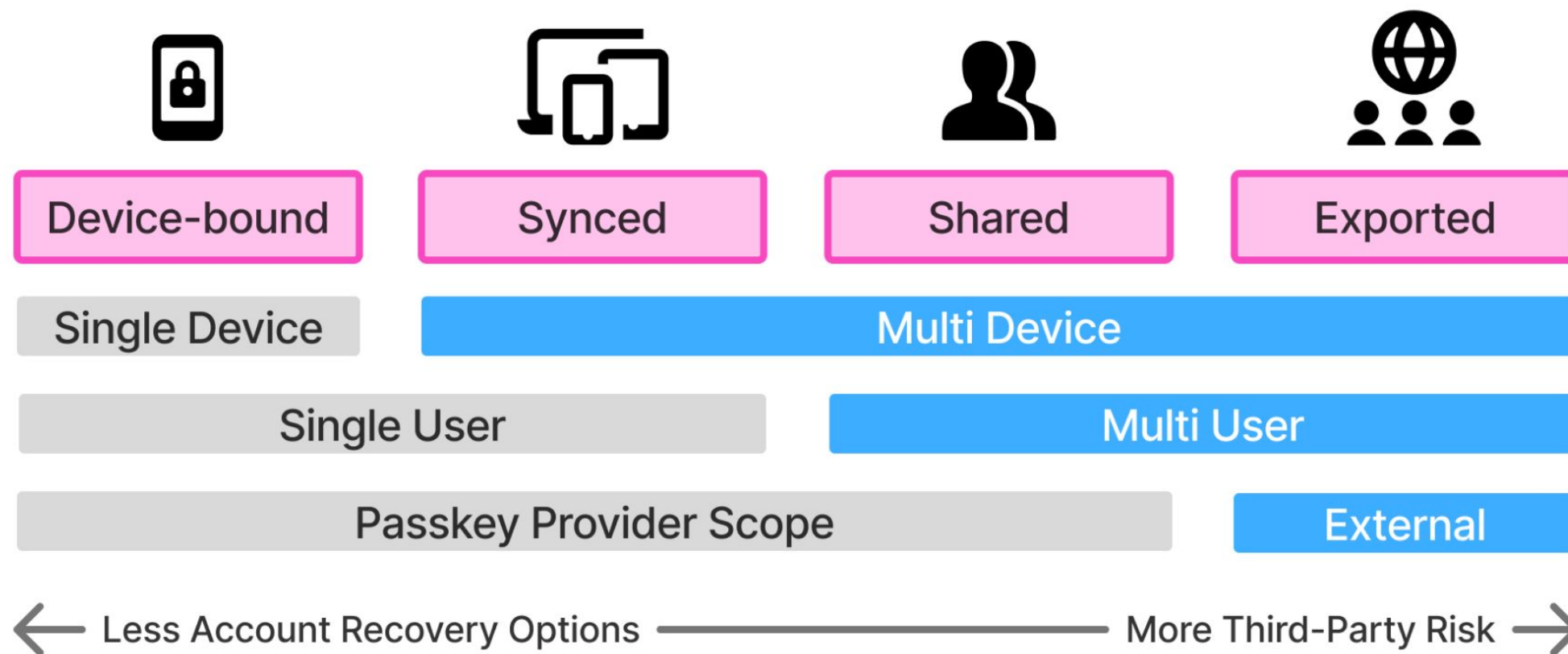
From login to keys

The PRF extension lets a passkey output a **32-byte secret** derived inside the authenticator.

The secret key **never leaves the device**. The app uses the output directly as a symmetric encryption key.



Access Levels of Passkey Sharing



SECTION 3 / 5

Design

A novel system bringing age and WebAuthn PRF together in a working prototype.



1 Problem

2 Background

3 Design

4 Study

5 Results

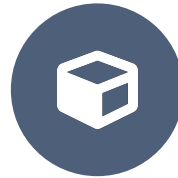
Three phases of design-based research



1. Framework Analysis

Build a comparative framework and systematically evaluate existing document encryption methods.

Chapter 4



2. System Design

Derive design requirements and iteratively build a prototype.

Chapter 5



3. User Study

Run a controlled, within-subjects study comparing the new system against password-based encryption.

Chapters 6 & 7

Designed & implemented system

age

per-file, multi-recipient,
open-source, PQ-ready encryption

+

FIDO2 passkeys

fairly familiar, non-custodial,
phishing-resistant identity

+

WebAuthn PRF

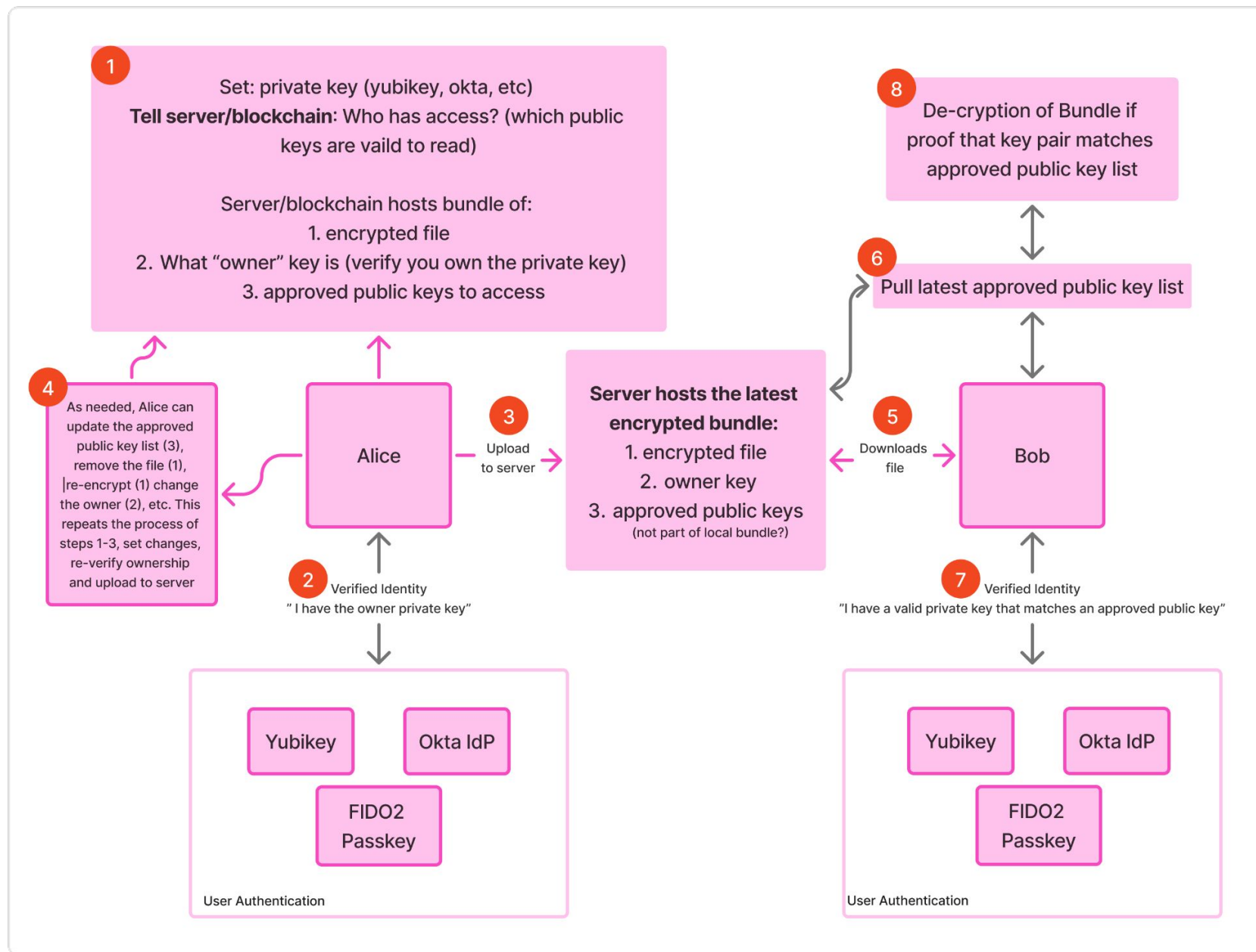
derives symmetric key from passkey,
for end-to-end encryption per file



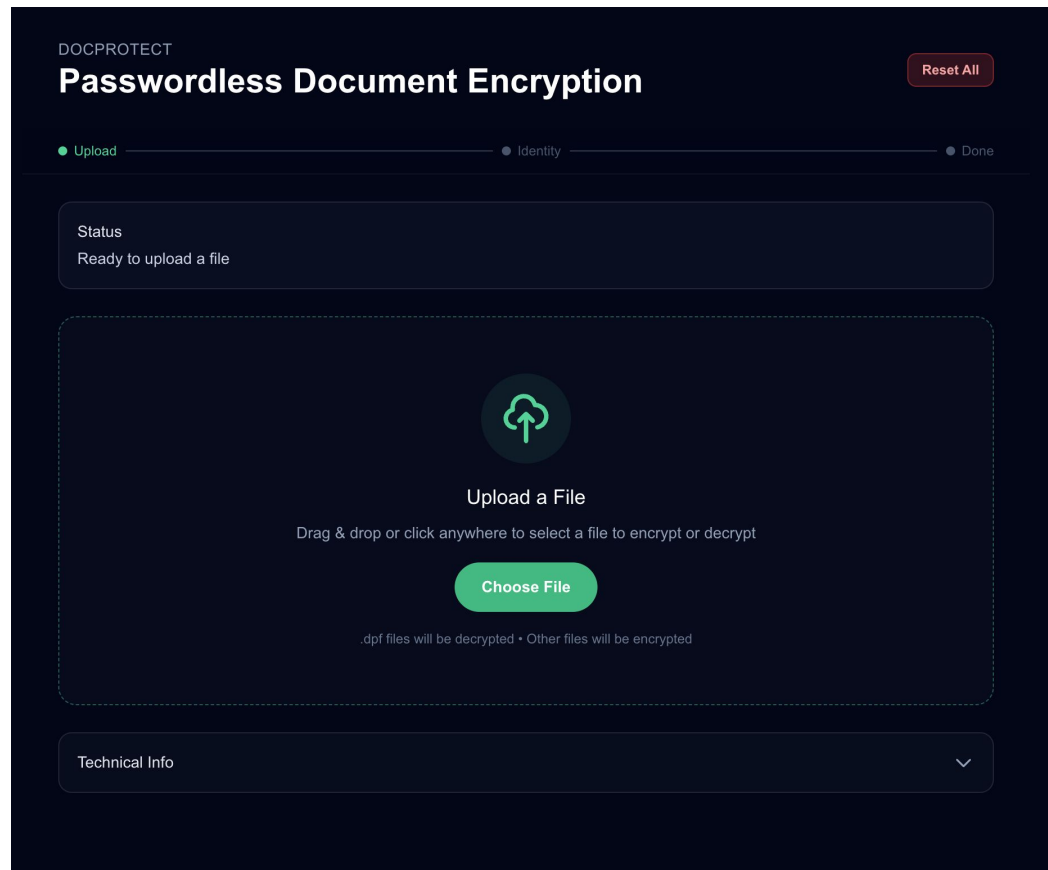
Passwordless Document Encryption

Non-custodial, passwordless, per-file end-to-end document encryption, entirely in the browser.
The server only holds ciphertext and public keys; encryption and keys never leave the client.





Functional prototype system



- 1 **Upload any file:** Encrypt or decrypt from a single browser interface. Nothing to install.
- 2 **Create or use a passkey:** Identity is verified with a fingerprint, face scan, or security key.
- 3 **Add recipients by email:** age multi-recipient encryption, bound to each recipient's passkey.
- 4 **Share a link or download:** file downloads for offline-sharing, creates realistic link imitates future features.

Implemented system evaluation

	Benefit	Rating
U1	Memorywise-Effortless	● Full
U2	Scalable-for-Users	● Full
U3	Easy-to-Learn	● Full
U4	Easy-Recovery-from-Loss	● Partial
U5	Seamless-Sharing	● Partial

	Benefit	Rating
D1	Negligible-Cost-per-User	● Full
D2	Negligible-IT-Support-Cost	● Full
D3	Compatible	● Partial
D4	Mature	○ Not satisfied
D5	Non-Proprietary	● Full
D6	Guest-Friendly	● Partial

	Benefit	Rating
S1	Resilient-to-Observation	● Full
S2	Resilient-to-Guessing	● Full
S3	Resilient-to-Third-Party-Leak	● Full
S4	Granular-Protection	● Full

SECTION 4 / 5

Study

Counterbalanced within-subjects evaluation of the new system against Adobe Acrobat (N = 21).



1 Problem

2 Background

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Within-subjects user study

N = 21

participants, each used
both systems

Within-subjects

counterbalanced order
(Passwordless ↔ Adobe Acrobat)

IRB-approved

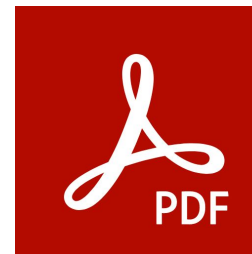
Exempt as low-risk study

Mixed methods

validated scales +
semi-structured interviews



VS.



Instruments

SUS

System Usability Scale: single 0–100 usability score, benchmarked

VdL

Van der Laan: usefulness & satisfaction acceptance subscales

ATI

Affinity for Technology Interaction: possible covariate

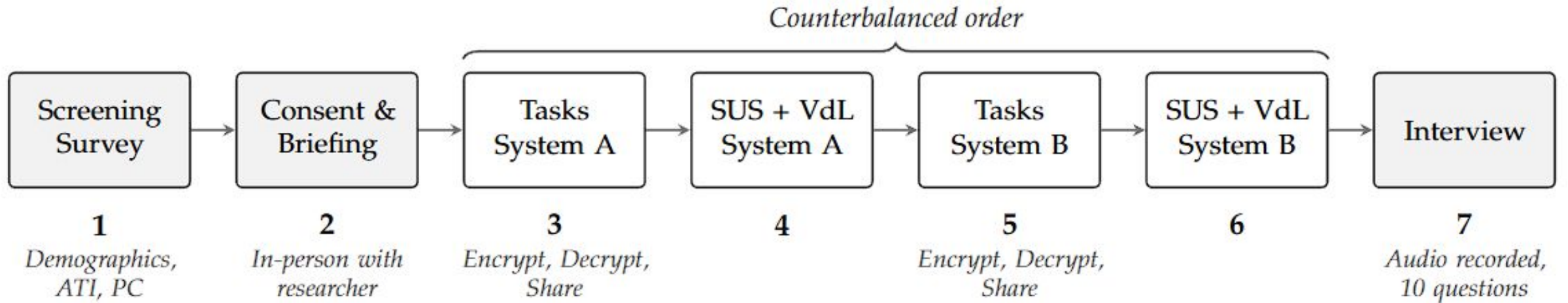
PC

Privacy Concern scale: possible covariate

Q+A

10 interview questions, thematic analysis

Within-subjects user study



vs.



Participants (N = 21)

29.2

mean age (SD 11.7)

71%

had a technical background

76%

male · 24% female

67%

held a bachelor's degree

ATI (technology affinity, 1–6): M = 4.11

Privacy Concern (1–7): M = 5.58

Scales nearly all $\alpha \geq .80$

Variable	Sample ($N = 21$)
Gender	
Male	76.2% (16)
Female	23.8% (5)
Age	$M = 29.2, SD = 11.7$
18–24	38.1% (8)
25–34	47.6% (10)
35–44	0.0% (0)
45–54	4.8% (1)
55–64	9.5% (2)
Education	
High school graduate	4.8% (1)
Some college	19.0% (4)
Bachelor's degree	66.7% (14)
Master's degree	4.8% (1)
Doctorate	4.8% (1)
Technical background	
Yes	71.4% (15)
No	28.6% (6)
ATI (1–6; $\alpha = .893$)	$M = 4.11, SD = 0.91$
Privacy Concerns (1–7; $\alpha = .803$)	$M = 5.58, SD = 1.01$

Internal consistency (Cronbach's alpha)

Instrument	Password-based	Passwordless
SUS (10 items)	.807	.887
VdL (9 items)	.884	.890
<i>Usefulness</i>	.646	.834
<i>Satisfaction</i>	.902	.813
ATI (9 items)	.893	
PC (4 items)	.803	

Note: ATI and PC were administered once per participant (not per condition).

SECTION 5 / 5

Results

Usability, acceptance, and adoption findings — plus takeaways and future research.



1 Problem

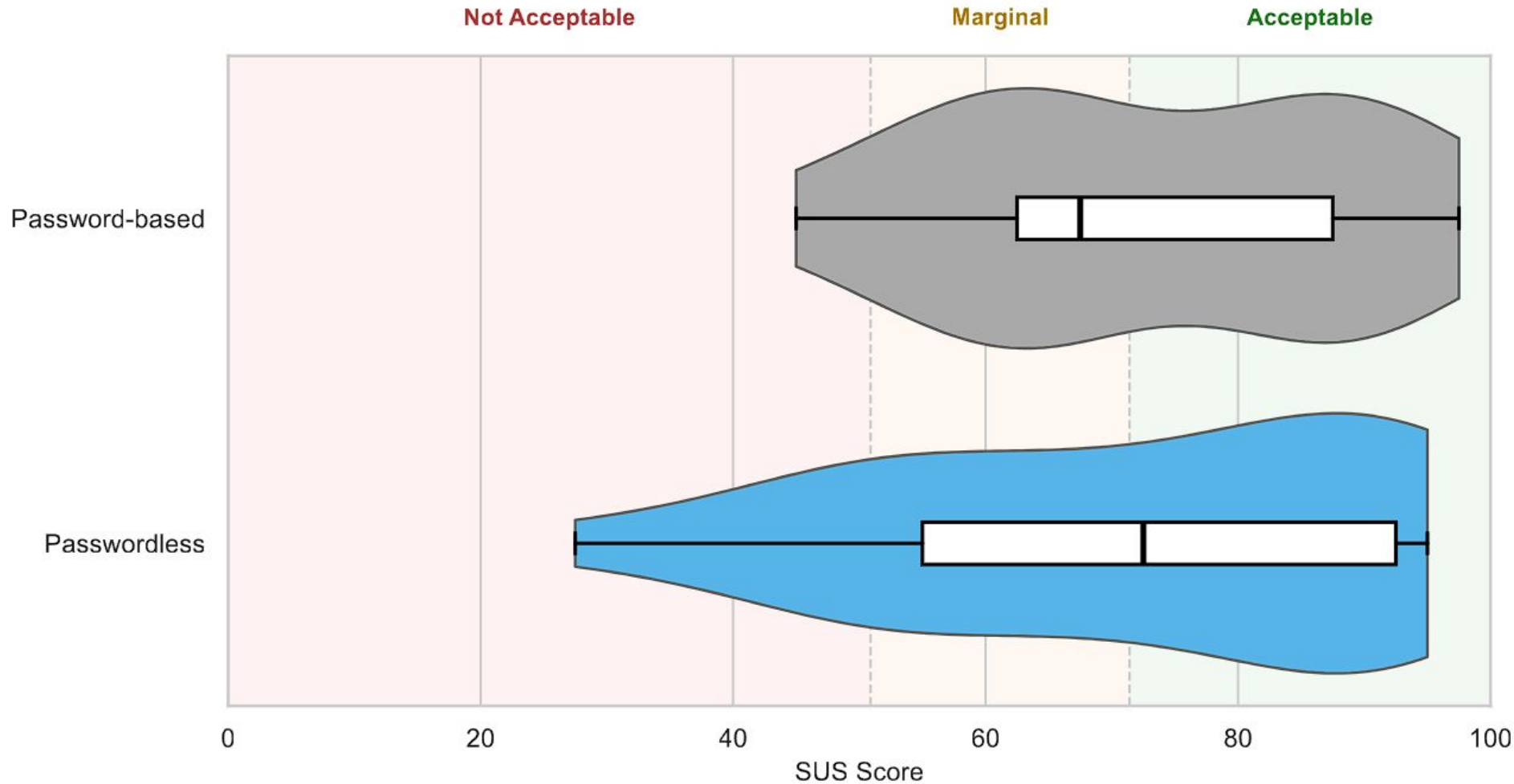
2 Background

3 Design

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Finding 1: usability parity on first use



SUS scores

73.3 password-based

72.0 passwordless

No significant difference
(Wilcoxon $W = 113.5$, $p = .944$).

A brand-new passwordless prototype matched a mature commercial tool on first use.

SUS benchmarking

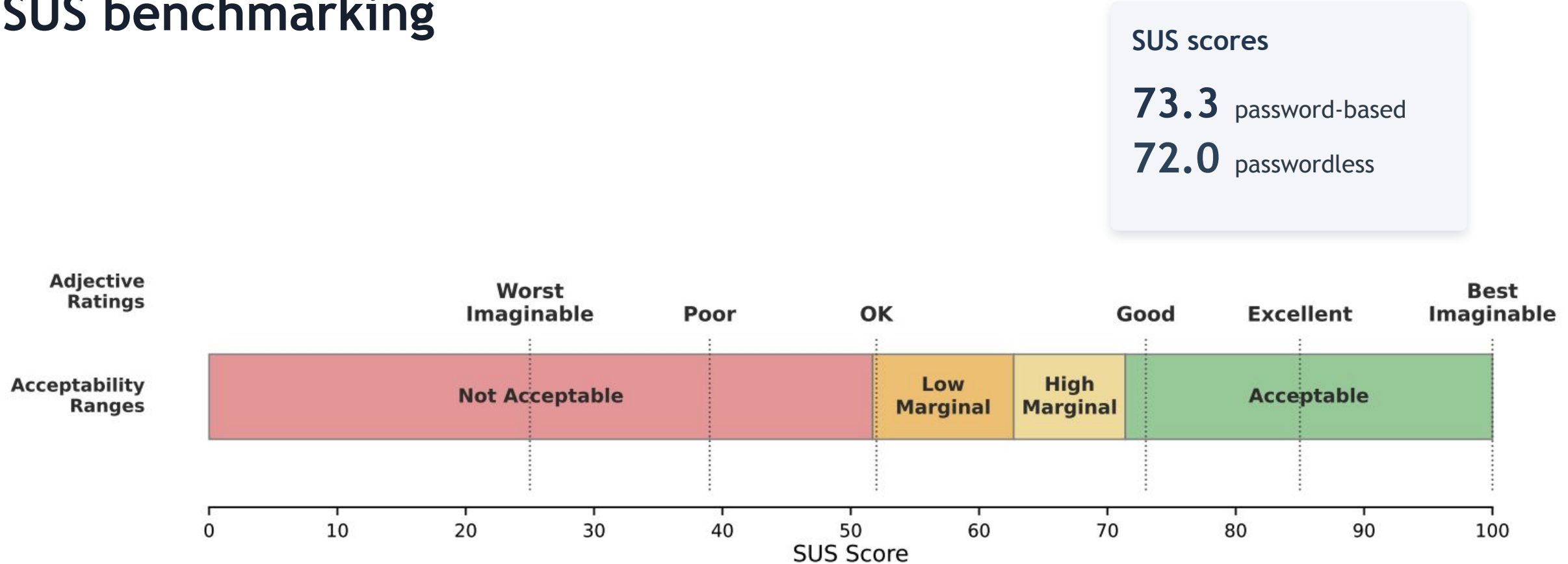
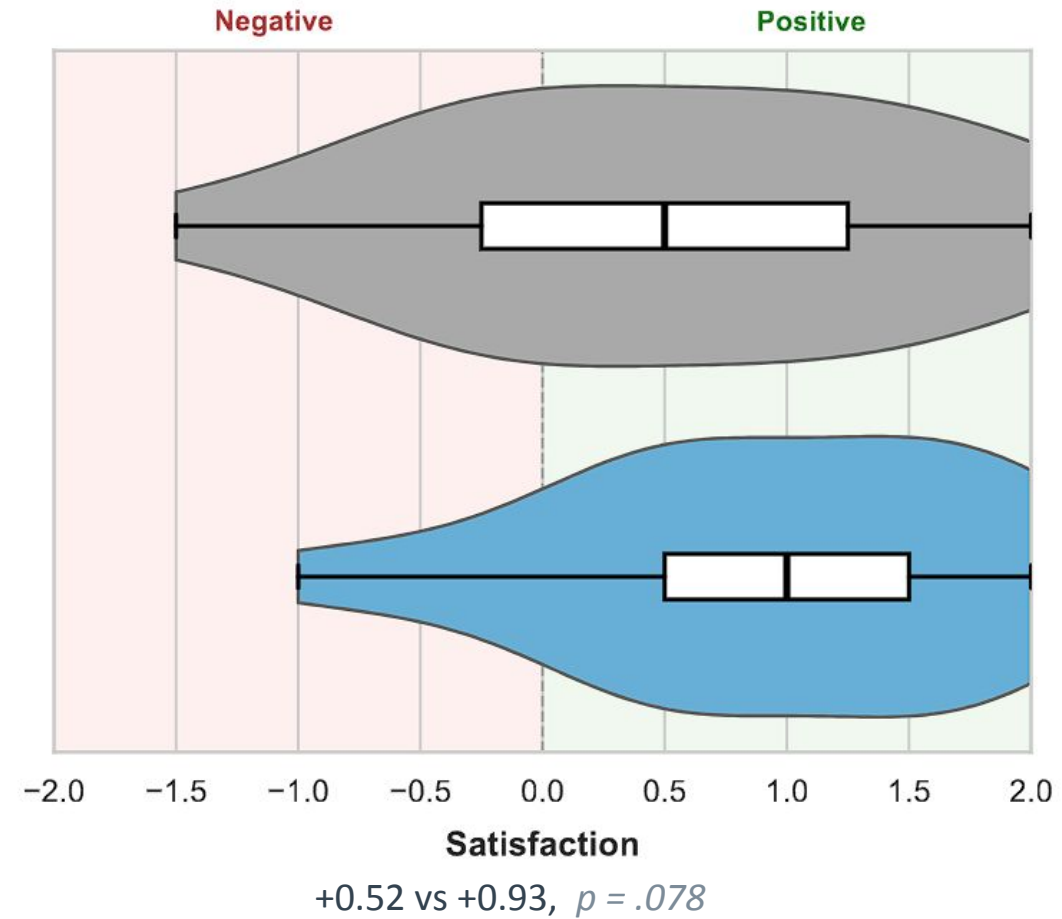
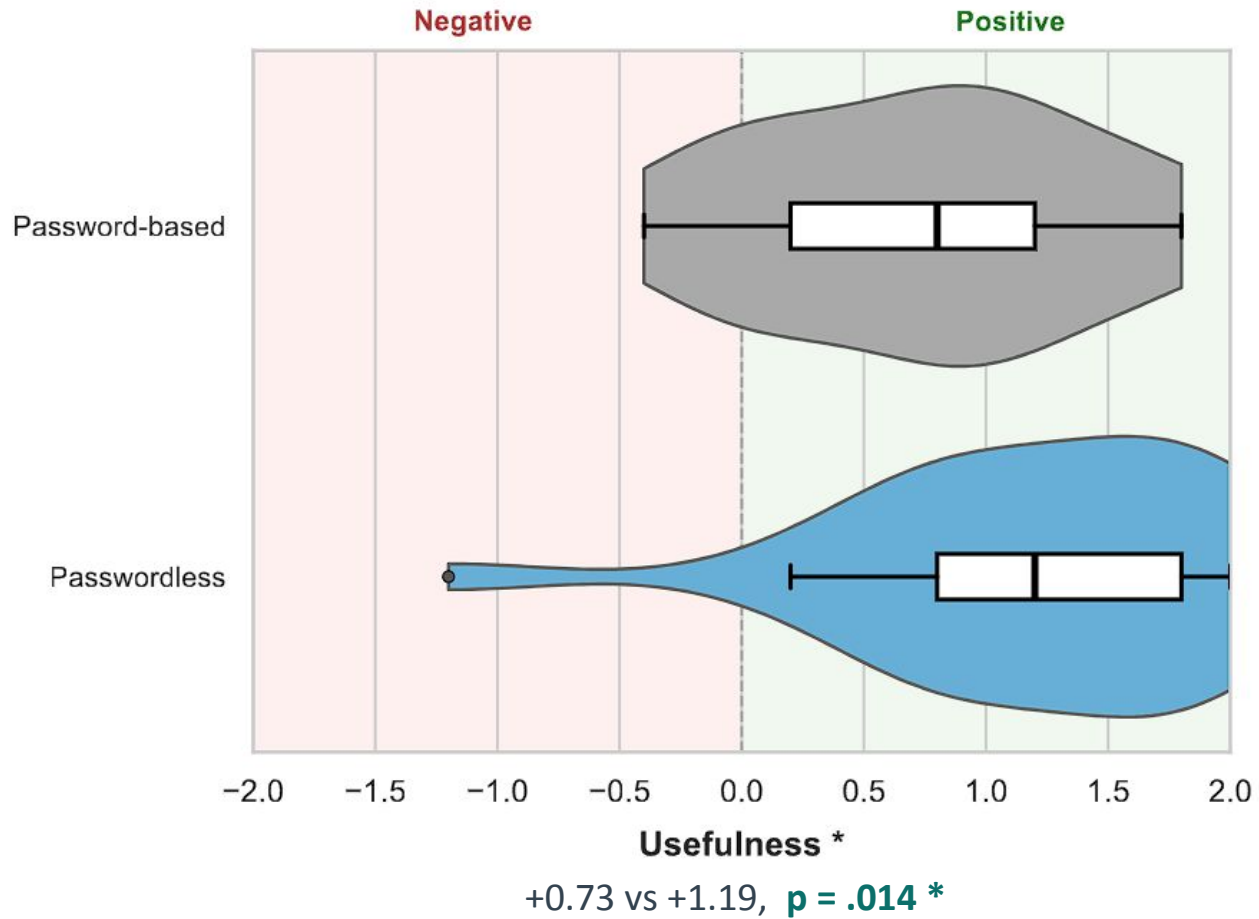
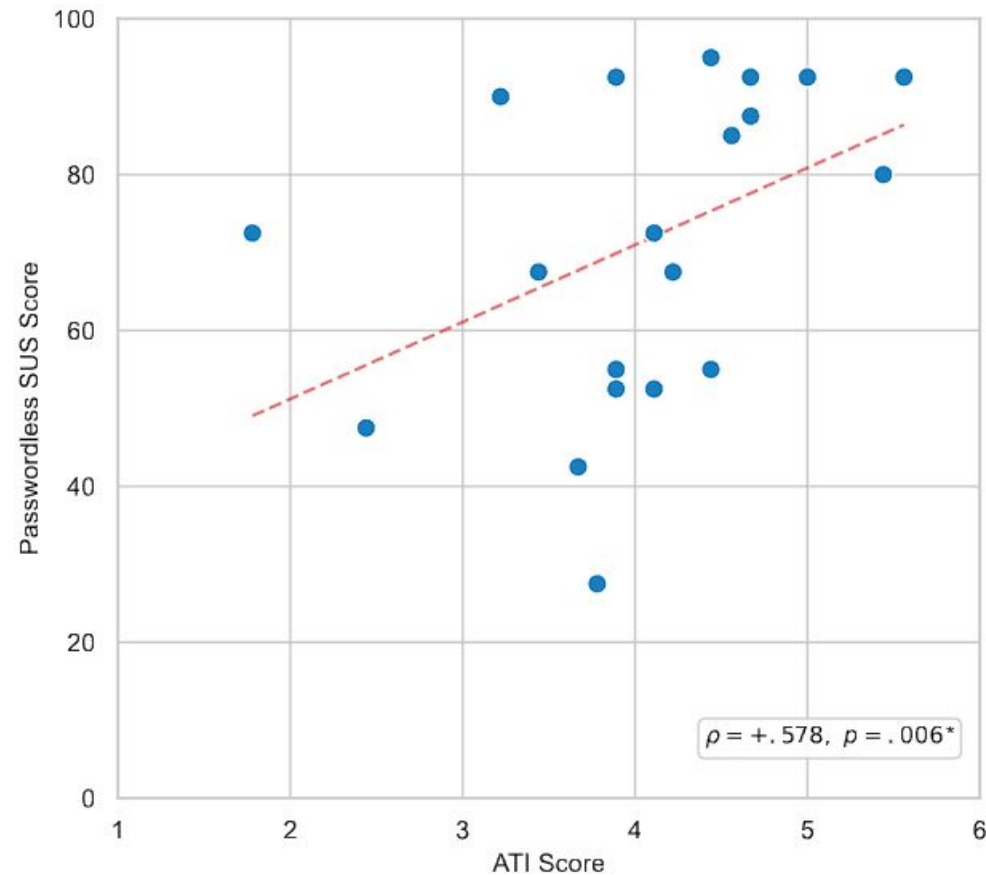
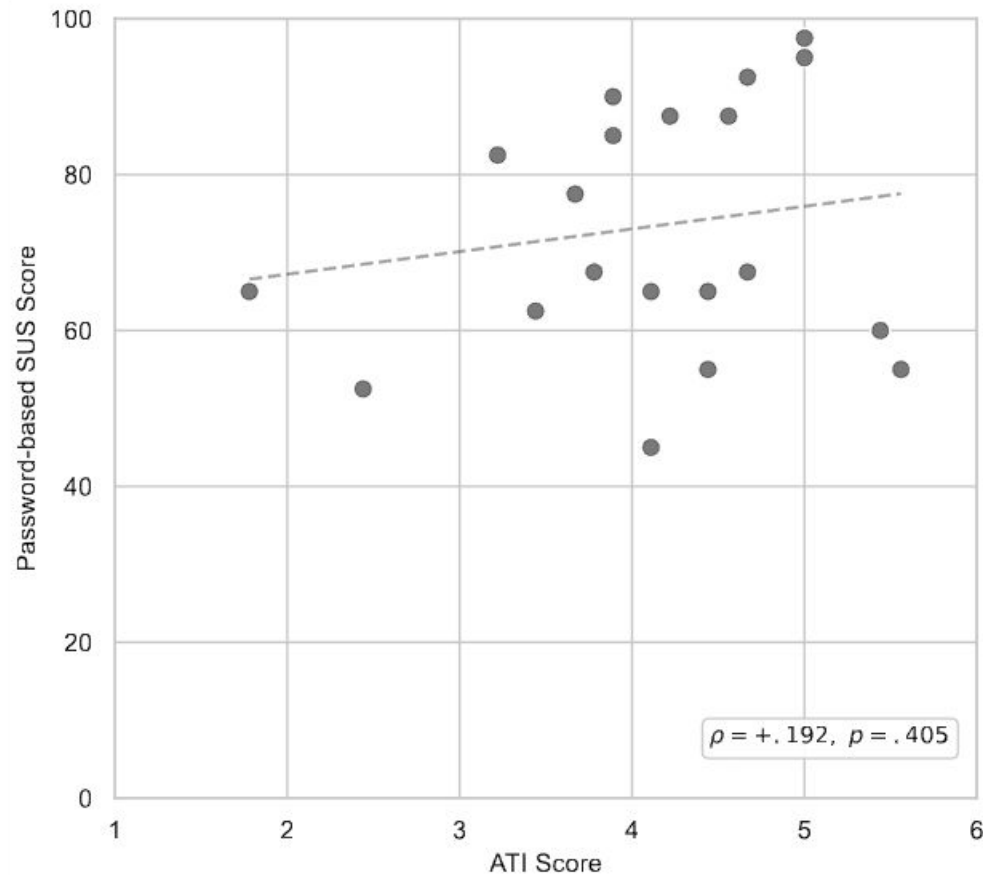


Figure 6.2: SUS score interpretation framework showing adjective ratings and acceptability ranges. Adapted from Bangor, Kortum, and Miller [84].

Finding 2: passwordless rated significantly more useful



Finding 3: tech affinity predicts perceived usability



$$\rho = +.578$$

($p = .006$)

Technology affinity (ATI) strongly predicts how usable people find the new passwordless system but does NOT predict ratings of the familiar password-based system.

Echoes Kupris & Schreck's (IEEE S&P 2026) ATI to usability finding.

Finding 3: tech affinity predicts perceived usability

Table 7.2: Spearman Correlations Between Control and Dependent Variables

Pair	ρ	p
ATI vs. Passwordless SUS	+.578	.006**
ATI vs. Password-based SUS	+.192	.405
ATI vs. Δ SUS	+.304	.180
ATI vs. Δ VdL Usefulness	+.367	.102
ATI vs. Δ VdL Satisfaction	+.168	.467
PC vs. Passwordless SUS	+.062	.788
PC vs. Δ VdL Usefulness	-.097	.676
ATI vs. PC	+.395	.077

Note: $N = 21$. Δ = Passwordless – Password-based. * $p < .05$, ** $p < .01$.

Finding 4: clear preference to adopt passwordless

21 / 21

willing to use the passwordless method
for their own documents today

19 / 21

preferred passwordless for sending
many documents at scale

(1 chose password, 1 torn)

“You do it once and you get it. I can see long-term the passkeys are better... it’s more organized.” — P24

Qualitative findings

1 Password sharing is the pain point

Most named securely sharing the password as the hardest part of the password-based method.

2 Trust in biometrics

Fingerprint / passkey verification gave a strong feeling of security.

3 Familiar to start, better at scale

Passwords felt simpler at first; passwordless won for repeated, real-world sharing.

4 Usability friction on first use

The unfamiliar passkey workflow added time for some. Various friction points with passkeys remain.

5 Adoption concerns

Recipients needing the tool, and limited understanding, were the main hesitations.

Contributions

1

Framework for comparison

15 benefits across usability, deployability, and security for evaluating any document encryption method.

2

Real-world grounding

Analysis of the MFT breach pattern that impacted millions. The work addresses a current, pressing problem.

3

Novel system

First application of FIDO2 + WebAuthn PRF to end-to-end document encryption, built as open-source browser based tool.

4

Empirical evidence

One of the first human-subjects usability evaluations of the PRF extension in any context.

Limitations & future work

Limitations

- Small, technically literate sample (N = 21) limits generalizability
- Single lab session measures first use, not long-term adoption
- Privacy-concern ceiling effect
- Prototype is not yet “mature” (D4) at production scale

Future work

- Larger, more diverse, non-technical participant samples
- Longitudinal study of repeated real-world use
- Onboarding to ease friction for lower-ATI users
- Recovery & key-portability process; production hardening

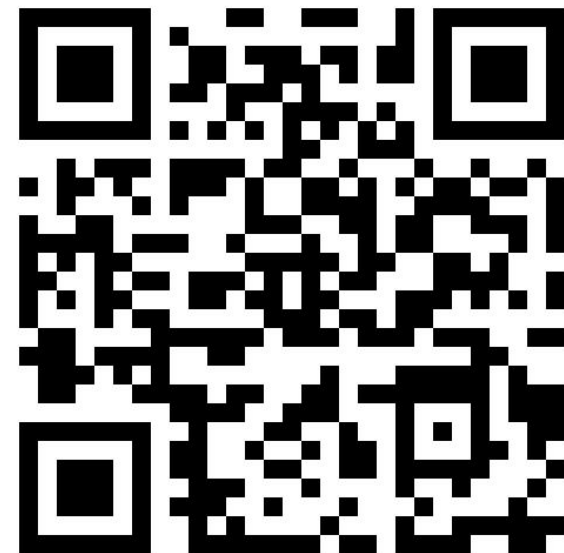
CONCLUSION

Passwordless document encryption is viable

- ✓ Framework showed no existing method satisfies all 15 benefits, designed system carries the most benefits
- ✓ Novel system uses age + FIDO2 + WebAuthn PRF to provide non-custodial, per-file, in-browser encryption
- ✓ On first use it matched Adobe on usability and was rated significantly more useful
- ✓ Users strongly preferred it for real-world document sharing
- ✓ Addresses the weaknesses that cause MFT breaches

Thank you!

Questions & discussion



Try the live demo

doc-protect.vercel.app